

Leakage-Safe Transfer Learning for Still-Image Posture Recognition

A locked POLAR study with external-transfer and attribution audits

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0.940 Macro-F1	0.946 Accuracy	3,329 Held-out images
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Abstract

This study evaluates ConvNeXt and DINOv2 transfer learning on a cleaned four-class POLAR subset. A source-aware audit quarantined 125 cross-split related images before fitting. All model, classifier, seed, epoch, and blend decisions were locked on 13,285 development images. Nine neural fits and three probes were hash-verified before one-time evaluation on 3,329 held-out images. The predeclared ensemble achieved 0.9399 macro-F1 (95% CI [0.9312, 0.9481]). External V-COCO transfer and attribution sanity show that this strong in-domain result is not equivalent to deployment readiness.

Scope: reproducible benchmark, not an exact state-of-the-art claim.

1. Study design and evidence boundary

The central design constraint is that the official test partition cannot influence model selection. The official train split fitted candidates; validation selected views, adaptation depth, regularization, classifier settings, epoch rules, and ensemble weights. Train and validation were combined only after the final lock.

- Primary task: sitting, standing, walking, and running; macro-F1 is primary.
- Secondary task: walking and running collapsed into one class.
- Uncertainty: 10,000 class-stratified bootstrap resamples, seed 20260822.
- Comparison: paired resampling on the same test rows.
- External and explanation audits have selection role 'none'.

1.1 Clean POLAR split

Split	Sitting	Standing	Walking	Running	Total
Train	3,021	2,720	1,994	2,223	9,958
Validation	1,000	967	648	712	3,327
Test	1,034	921	638	736	3,329

1.2 Source-related quarantine

Byte hashes found no exact cross-split duplicates, but perceptual retrieval identified alternate crops, color variants, and nearby source frames. Confirmation required a 64-bit perceptual-hash distance at most six and normalized grayscale correlation at least 0.90, or equivalent independent source evidence. The audit quarantined 125 images in 61 connected components before supervised fitting. No test image moved into development.

Control	Recorded value
Clean development rows	13,285
Verified neural fits	9 (3 families x 3 seeds)
Verified frozen-feature probes	3
Final-fit test rows read	0
Official test-manifest opens	1
Selection-lock SHA-256	fa3fb7c80a073d29048afb8e0b8da1fb17f5ade9721630347c37523714cca187

2. Models and final-stage classifiers

Component	View	Adaptation	Augmentation	Dropout	Epochs
ConvNeXt-S	Full frame	Full; layer decay 0.70	Mild	0.10	12
DINOv2-S	Person + 25% context	Full; layer decay 0.75	Moderate	0.10	12
DINOv2-B	Person + 25% context	Top four blocks	Mild	0.10	7

Neural components average seeds 42, 52, and 62. Fixed epoch counts are the median best epochs from corresponding development confirmation runs. The strongest frozen representation concatenates normalized CLS tokens from the last four DINOv2-B layers and the final mean patch token across full-frame and person-plus-10%-context views, yielding 7,680 features.

2.1 Classifier probes

The linear endpoint is standardized multinomial logistic regression with $C=0.001$. The nonlinear endpoint is an RBF SVM with $C=10$ and $\gamma=1/7680$, followed by five-fold sigmoid calibration. RBF settings were transferred from the declared development screen rather than retuned on final features.

2.2 Locked probability blend

Component	Weight
ConvNeXt-S full	20%
DINOv2-B logistic	20%
DINOv2-B RBF SVM	20%
DINOv2-B top four	25%
DINOv2-S full	15%

All final fits completed in 3.21 hours. Checkpoint and pipeline hashes were verified before the test gate opened.

3. Locked held-out results

Candidate	Macro-F1	95% CI	Accuracy	Log loss	ECE
Locked ensemble	0.9399	[0.9312, 0.9481]	0.9456	0.1564	0.0291
DINOV2-B RBF SVM	0.9274	[0.9179, 0.9362]	0.9342	0.2280	0.0269
DINOV2-B logistic	0.9258	[0.9164, 0.9348]	0.9324	0.1764	0.0133
DINOV2-B top four	0.9252	[0.9156, 0.9343]	0.9327	0.2173	0.0420
DINOV2-S full	0.9131	[0.9028, 0.9226]	0.9210	0.2389	0.0298
ConvNeXt-S full	0.8914	[0.8803, 0.9020]	0.8994	0.3081	0.0429

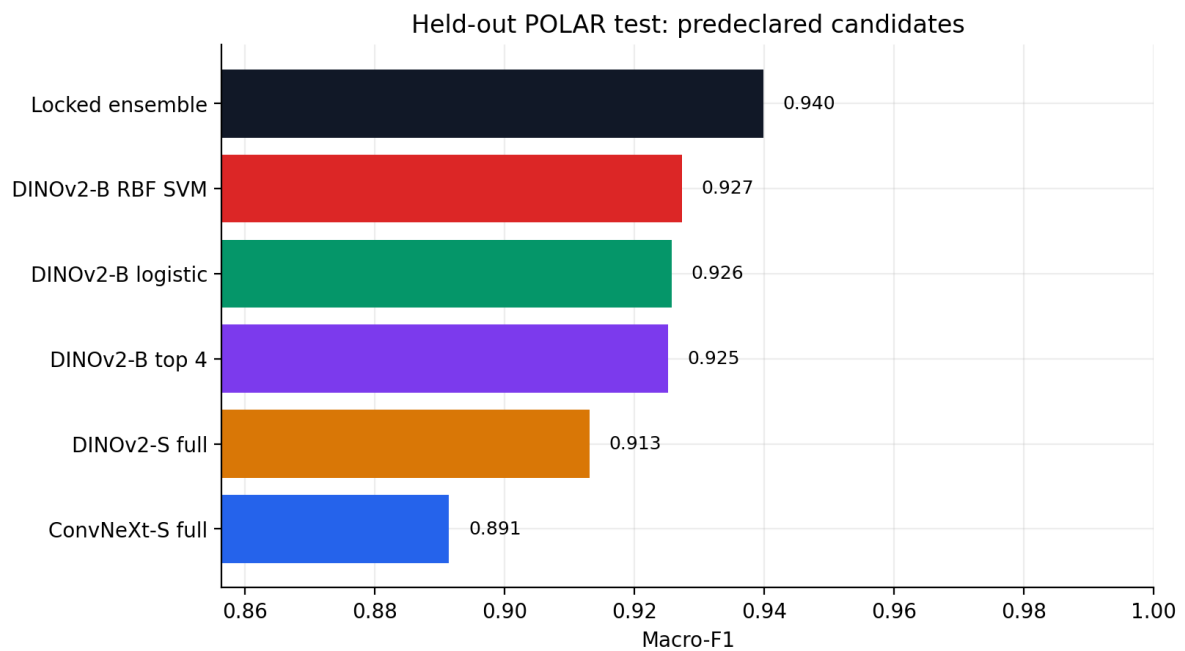


Figure 1. Every candidate was declared before test access. The ensemble is the primary candidate regardless of the observed ordering.

The ensemble's paired 95% macro-F1 interval is positive against every component. The smallest gain is +0.0125, with interval [0.0065, 0.0186]. The collapsed three-class ensemble reaches 0.9611 macro-F1 and 0.9622 accuracy.

4. Class behavior

Class	Precision	Recall	F1	Support
Running	0.957	0.961	0.959	736
Sitting	0.992	0.985	0.989	1,034
Standing	0.925	0.939	0.932	921
Walking	0.887	0.873	0.880	638

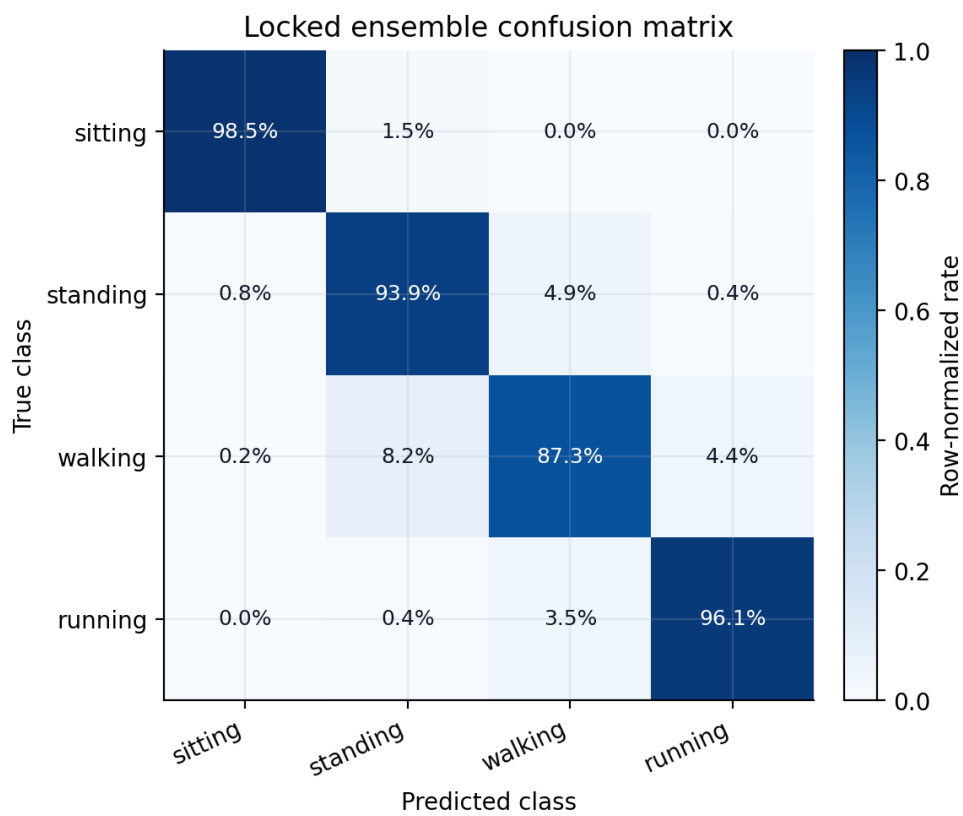


Figure 2. Row-normalized confusion for the locked ensemble. Walking is hardest: 8.2% of walking rows are predicted as standing and 4.4% as running.

5. Where the gain came from

The frozen DINOv2-B curve rises from 0.8487 macro-F1 at 242 training images to 0.9150 at 9,958. This 6.64-point increase is larger than any single seed-42 regularization intervention.

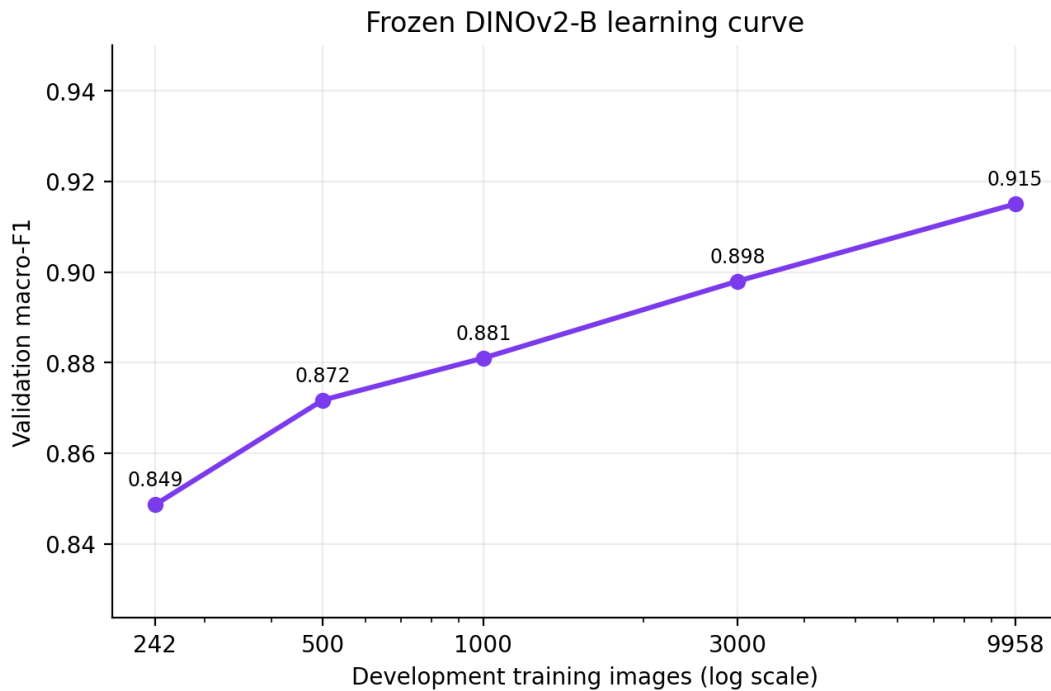


Figure 3. Frozen DINOv2-B validation learning curve.

5.1 Regularization screen

Seed-42 DINOv2-S intervention	Validation macro-F1
Dropout 0.20	0.9191
Dropout 0.10 + mild augmentation	0.9184
Moderate augmentation	0.9182
Label smoothing 0.05	0.9142
No random erasing	0.9124
MixUp 0.20	0.9120
No dropout	0.9113
Inverse-frequency weights	0.9082

The evidence supports retaining dropout and image augmentation, not stacking every available regularizer. Multi-seed stability and complementarity favored the moderate DINOv2-S variant for the blend.

6. SVM tradeoff and external transfer

Probe	Test F1	Log loss	External F1	Fit time	Size
Logistic	0.9258	0.1764	0.6392	13.9 s	0.4 MB
Calibrated RBF SVM	0.9274	0.2280	0.6504	60.4 min	870.9 MB

The nonlinear boundary adds a small accuracy gain and useful ensemble diversity, but logistic regression is the practical endpoint for compact, calibrated serving. The RBF model averages 5038 support vectors per calibration fold and is more than 2,000 times larger.

6.1 V-COCO no-retuning evaluation

The overlap audit compared 16,614 clean POLAR records with 4,123 V-COCO images and found zero exact matches, zero perceptual candidates, and zero confirmed source-related pairs. The locked models were evaluated without retuning.

Candidate	Macro-F1	Accuracy	Log loss
DINOv2-B top four	0.6732	0.6751	1.7509
Locked ensemble	0.6669	0.6663	1.2512
DINOv2-B RBF SVM	0.6504	0.6453	2.0801
DINOv2-S full	0.6455	0.6472	1.6232
Direct three-class probe	0.6430	0.6405	1.2775
DINOv2-B logistic	0.6392	0.6363	1.4022
ConvNeXt-S full	0.6208	0.6272	1.8414

7. External result and interpretation

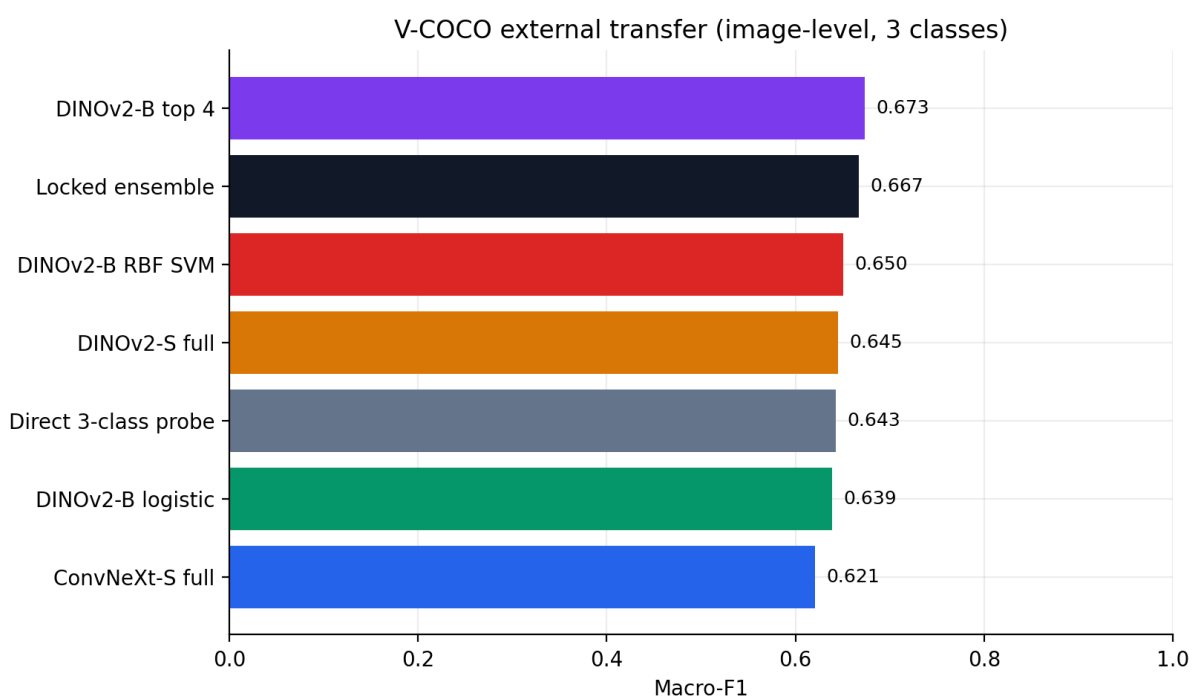


Figure 4. Image-level V-COCO macro-F1 for fixed three-class mappings. DINOv2-B top-four adaptation transfers best descriptively; the locked ensemble remains the predeclared primary candidate.

The collapsed ensemble falls from 0.9611 in-domain macro-F1 to 0.6669 externally, a 29.4-point gap. External standing recall is 0.412, while walking/running recall is 0.929 but precision is 0.349. Image composition and annotation semantics therefore change the decision problem. The result argues for domain-generalization work rather than another POLAR-only tuning loop.

- In-domain selection and external robustness are not aligned objectives.
- The adapted DINOv2-B component transfers better than the in-domain-optimal blend.
- External results diagnose generalization; they do not reopen model selection.

8. Attribution faithfulness

The fixed cohort contains 256 images balanced by class and person-box-area quartile. ConvNeXt uses Grad-CAM and DINOv2-B uses 16-step integrated gradients. The audit combines deletion/insertion, nested random deletion, bbox localization, equal-area person/context occlusion, full/crop consistency, alternative targets, and two levels of parameter randomization.

Metric	ConvNeXt-S	DINOv2-B
Deletion AUC (lower)	0.615	0.874
Random-minus-targeted gap	0.163	0.058
Insertion AUC	0.896	0.946
Person mass / bbox area	2.368	1.095
Person-minus-context drop	0.234	0.012
Alternative-target Spearman	-0.491	0.934
Randomized-head Spearman	0.183	0.871
Randomized-layers Spearman	0.135	0.708

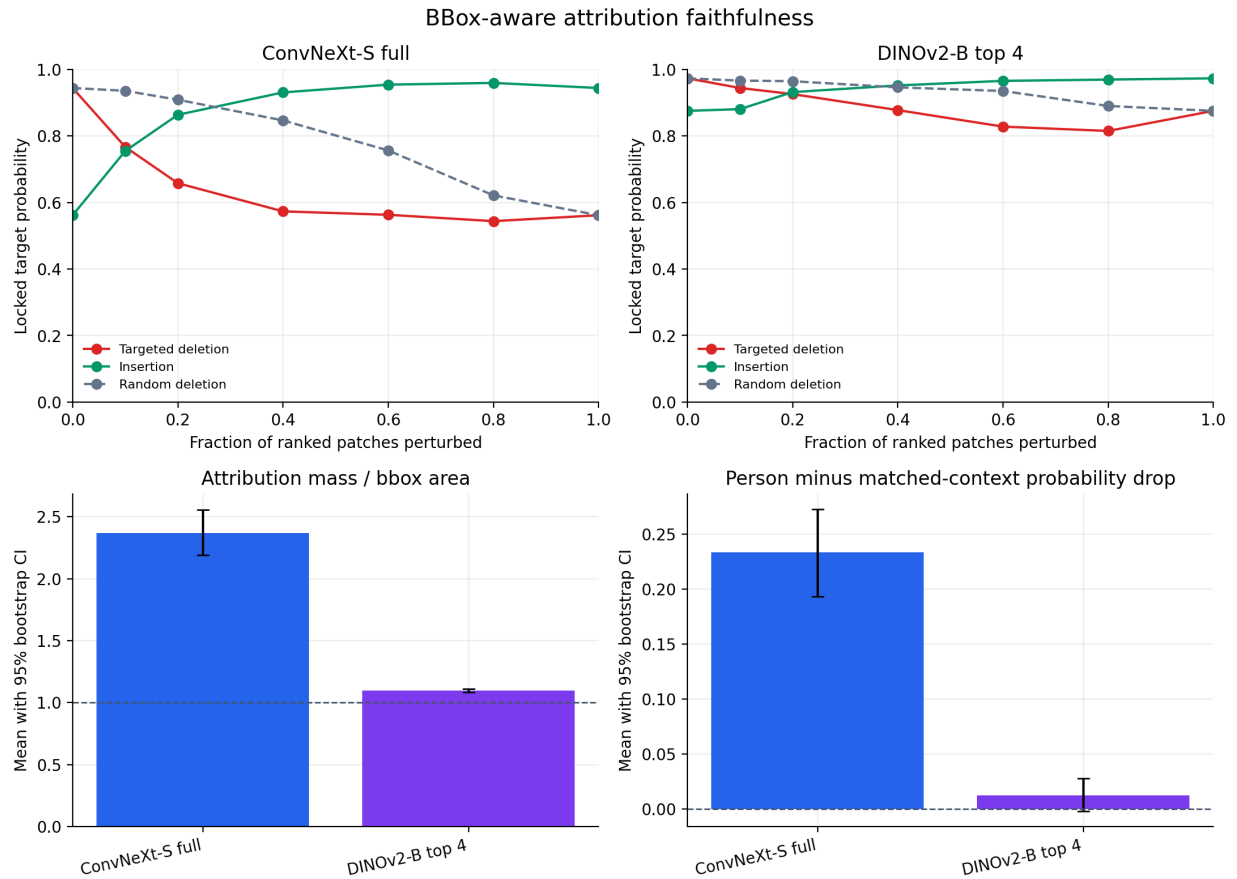


Figure 5. Perturbation curves and bbox-aware attribution diagnostics.

9. Attribution sanity result

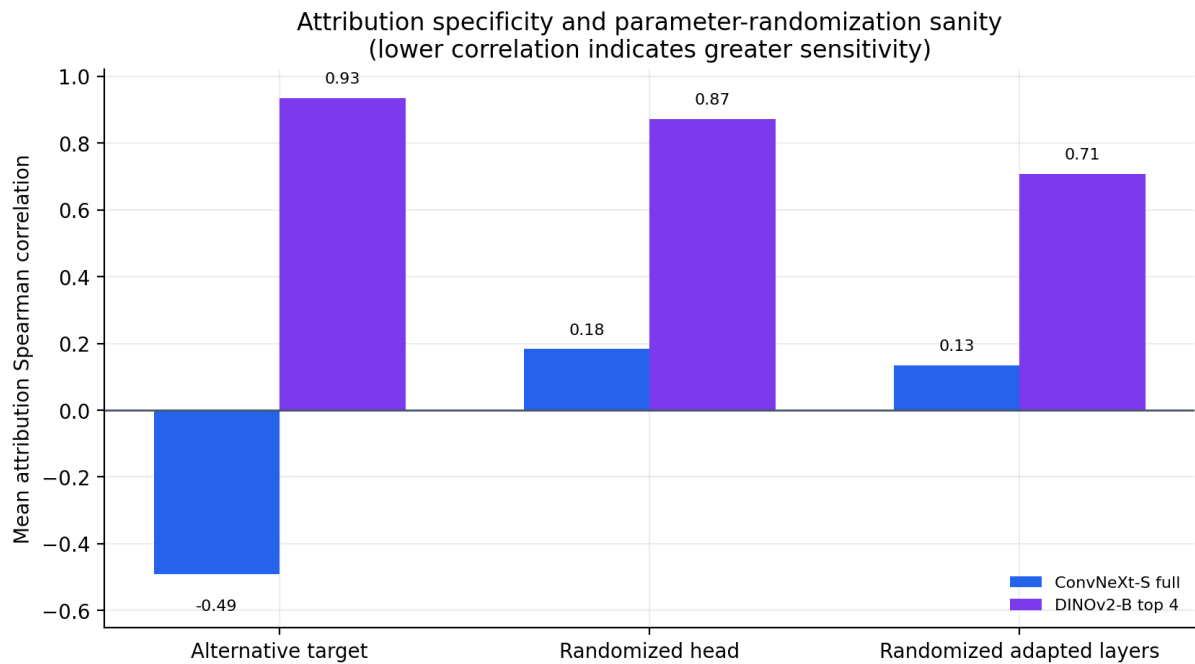


Figure 6. Lower correlation after changing the target or learned parameters indicates greater explanation sensitivity.

ConvNeXt provides the stronger evidence: targeted deletion separates from random deletion, person attribution has 2.37x area-normalized lift, and maps change after target and parameter randomization. DINOv2-B integrated gradients place 84.8% of positive mass inside person boxes, but area-normalized lift is only 1.10 and maps remain highly correlated after both sanity interventions. DINO integrated gradients are therefore retained as coarse localization diagnostics, not promoted as faithful causal explanations.

Two rows per family had full-frame projected person boxes and no available equal-area context. They remain in every other metric and are explicitly excluded only from the matched-context statistic. Maximum probability replay error was 0.00176.

10. Bounded bit-flip robustness

Input faults flip exact bits in post-resize uint8 RGB tensors before normalization. Parameter faults symmetrically quantize each classifier weight matrix to int8 and flip exact bits. Three declared fault seeds are averaged. This audit is reported separately from attribution faithfulness.

Condition	Level	ConvNeXt agreement	DINOv2 agreement
Input bit rate	1e-05	0.996	1.000
Input bit rate	0.0001	0.996	0.996
Input bit rate	0.001	0.984	0.980
Head bit flips	16	1.000	1.000

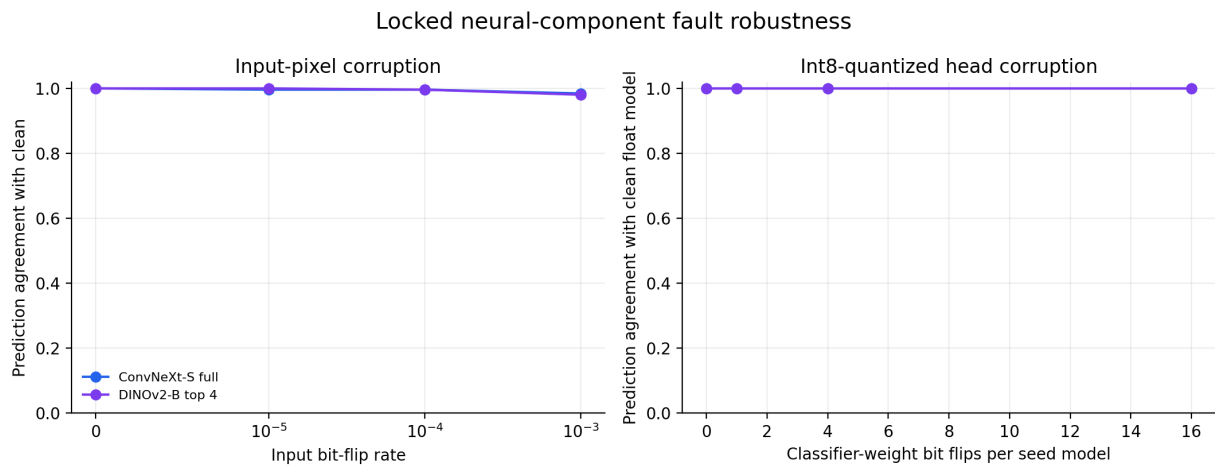


Figure 7. Agreement with clean predictions under declared fault levels.

The result shows local prediction stability on the fixed cohort. It does not test persistent memory faults, random backbone corruption, or hardware failure, and it is not a safety certification.

11. Conclusions and limitations

- Data scale is the largest isolated performance amplifier in the study.
- DINOv2-B representations support strong linear and nonlinear final-stage classifiers.
- The development-locked blend has a statistically supported gain over every component.
- The external-domain gap is large; POLAR performance is not a deployment guarantee.
- ConvNeXt Grad-CAM passes declared sanity checks more convincingly than DINOv2-B integrated gradients.

11.1 Claim boundary

No exact state-of-the-art claim is made. The task is a cleaned four-class subset rather than the full nine-label POLAR benchmark, and subject/session identifiers are not available. V-COCO uses a different source and annotation policy. A formal preprint would be stronger after replication on additional labels or an independent dataset with subject/session boundaries and a predeclared domain-generalization intervention.

11.2 Reproducibility

The repository publishes code, protocols, selection locks, exclusion records, metrics, uncertainty, figure builders, and per-file hashes. Raw images, checkpoints, fitted classifier binaries, local paths, and dense predictions remain local. GitHub Actions runs lint, compilation, tests, and the portable evidence validator on Linux.

Recommended publication form: technical report plus reproducible portfolio article. The contribution is the evidence discipline around a strong benchmark, including the negative external and attribution findings.

References

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Artifact availability

Repository: <https://github.com/abdullahuseyinli-dot/human-activity-classification>

Code and documentation license: MIT. Dataset images, annotations, and pretrained weights retain upstream terms.